

Geometric series



- 1 Find the sum of the first 5 terms of the series: $24 - 12 + 6 \dots$
Leave your answer correct to the nearest whole number.
- 2 Find the sum of the first 7 terms of the geometric series: $64 + 16 + 4 \dots$
- 3 Find the sum of the first 7 terms of the series: $8 + 1 + \frac{1}{8} \dots$
Round your answer to three decimal places.
- 4 Find the sum of the first 11 terms of the series: $1 - 2 + 4 \dots$
- 5 Find the sum of the first 12 terms of the series: $5 + 10 + 20 \dots$
- 6 Find the sum of the first 5 terms of the geometric sequence defined by the following. Round your answers to two decimal places.
 - a $u_1 = 2.187$ and $r = 1.134$
 - b $u_1 = -4.186$ and $r = -2.848$
- 7 For each of the following series:
 - i Find n , the number of terms in the series.
 - ii Find the sum of the series.
 - a $4 - 8 + 16 - \dots - 2048$
 - b $64 + 16 + 4 + \dots + \frac{1}{16}$
- 8 Find n , the number of terms, in each of the following series:
 - a The sum of n terms in the geometric series $2 + 10 + 50 + \dots$ is 195 312.
 - b The sum of n terms in the geometric series $5, -20, 80, \dots$ is 262 145.
 - c The sum of n terms in the geometric series $16 + 4 + 1 + \dots$ is $\frac{1365}{64}$.
 - d The sum of n terms in the geometric series $40, -8, \frac{8}{5}, \dots$ is $\frac{20\,832}{625}$.
- 9 Consider the series: $5 + \frac{5}{2} + \frac{5}{4} \dots$
 - a Find the common ratio, r .
 - b Find the sum of the first 10 terms, rounding your answer to one decimal place.

- 10 The first 3 terms of a sequence are x , $\frac{2}{5}x^2$, $\frac{1}{25}x^3$.
- a Write a simplified expression for the sum of the first n terms of this sequence.
- b Evaluate the sum when $x = 5$ and $n = 9$.
- 11 Consider the series: $\frac{1}{4} - \frac{1}{5} + \frac{1}{16} - \frac{1}{25} + \frac{1}{64} - \frac{1}{125} + \dots$
Form an expression for the sum of the first $2n$ terms of the series.
- 12 The sum of the first 6 terms of a geometric series is 28 times the sum of its first 3 terms.
Find r , the common ratio.
- 13 The sum of the first 9 terms of a geometric series is 436 905, and the common ratio is $\frac{1}{4}$.
Find u_1 , the first term in the series.

Summation notation

- 14 Write the following series using summation notation:

a $1 + 2 + 4 + \dots + 512$

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{1}{1024}$

c $-3 + 9 - 27 + \dots + 729$

d $6 + 18 + 54 + \dots + 486$

e $\frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \dots + \frac{1}{1024}$

f $8 + 4 + 2 + \dots + \frac{1}{8}$

- 15 Evaluate:

a $\sum_{i=1}^4 4^i$

b $\sum_{j=1}^5 243 \left(\frac{5}{3}\right)^j$

c $\sum_{j=1}^5 32 \left(\frac{1}{2}\right)^j$

d $\sum_{i=3}^8 (-3)^i$

e $\sum_{i=2}^7 (-2^i)$

f $\sum_{i=1}^5 (2(3)^{i-1})$

g $\sum_{j=4}^8 \left(\frac{1}{4}(3)^{j-1}\right)$

h $\sum_{j=1}^4 \left(-3\left(\frac{1}{3}\right)^j\right)$

- 16 Write the following series using summation notation:

$$x + 2x^4 + 4x^7 + 8x^{10} + \dots + 2^{18}x^{55}$$



Technology

- 17** Use the sequence facility of your calculator to determine for which term number n the geometric sequence $u_n = 6(2.1^{n-1})$ has a sum of 217.3146.
- 18** Use the sequence facility of your calculator to determine for which term number n the geometric sequence $u_n = 800(0.9^{n-1})$ has a sum of 4556.262 32.
- 19** The 5th term of a geometric sequence is 405. The sum of the 6th, 7th and 8th terms is 15 795.
- a** Write an equation involving u_1 , the first term, and r , the common ratio, of this geometric sequence for the 5th term.
 - b** Write an equation involving u_1 , the first term, and r , the common ratio, and the sum of the 6th, 7th and 8th terms of the sequence.
 - c** Use the simultaneous solving facility of your calculator to find the values of u_1 and r . Assume u_1 and r are positive.
 - d** Hence, state the general term, u_n , of the sequence.

Applications

- 20** Suppose you save \$1 the first day of a month, \$2 the second day, \$4 the third day, \$8 the fourth day, and so on. That is, each day you save twice as much as you did the day before.
- a** How much will your total savings be for the first 13 days?
 - b** How much will your total savings be for the first 29 days?
- 21** Average annual salaries are expected to increase by 5% each year. The average annual salary this year is found to be \$49 000.
- a** Calculate the expected average annual salary in 4 years.
 - b** This year, Aaron starts at a new job in which he will receive the average annual salary for each year of his employment. Over the coming 4 years (including this year), he plans to save half of each year's annual salary. Determine his total savings over these 4 years.

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- 22** This year, 600 people are expected to enter the workforce as registered nurses. This number is expected to increase by 4% next year, and increase by the same percentage every year after that.
Calculate the following, rounding your answers to the nearest whole number:
- a** The number of nurses expected to enter the workforce between six and seven years from now.
 - b** The number of nurses expected to enter the workforce over the next six years.
- 23** A conveyor belt is being used to remove materials from a quarry. Every thirty minutes, the conveyor belt empties out $\frac{1}{5}$ of whatever material remains in the quarry.
The quarry initially holds $14\,000\text{ m}^3$ of materials.
- a** How much material has been taken out after 90 minutes?
 - b** How much material is left in the quarry after 90 minutes?
- 24** The first blow of a hammer drives a post a distance of 64 cm into the ground. Each successive blow drives the post $\frac{3}{4}$ as far as the preceding blow. In order for the post to become stable, it needs to be driven $\frac{781}{4}$ cm into the ground.
If n is the number of hammer strikes needed for the pole to become stable, find n .
- 25** A car's brakes failed and the driver immediately turned the engine of the car off. In the first second after the engine was shut off, the car travelled 20 m. Every successive second, the car travelled 80% of the distance covered in the previous second.
- a** Find an expression for the total distance the car travelled in the first n seconds
 - b** Hence, find the total distance the car travelled in the first four seconds to two decimal places.
 - c** Traffic had built up 109 m away from where the driver turned off the engine.
How far from the traffic build-up did the car come to rest?

- 26** At the start of 2014, Pauline deposits \$5000 in an investment account. At the end of each quarter, she makes an extra deposit of \$700.

The table below shows the first few quarters of 2014. All values in the table are in dollars:

Quarter	Opening Balance	Interest	Deposit	Closing Balance
Jan - Mar	5000	200	700	5900
Apr - Jun	5900	236.00	700	6836.00
Jul - Sep	6836.00	273.44	700	7809.44

- Find the quarterly interest rate.
 - Write an expression for the amount in the account at the end of the first quarter.
 - Hence, write an expression for the amount in the account at the end of the second quarter.
 - Hence, write a similar expression for the amount in the account at the end of the third quarter.
 - Write an expression for the amount in the investment account after n quarters.
 - Hence, determine the total amount in Pauline's account at the beginning of 2016 to the nearest dollar.
- 27** Iain opens an account to help save for a house. He opens the account at the beginning of 2017 with an initial deposit of \$50 000 that is compounded annually at a rate of 5.9% per annum. He makes further deposits of \$3000 at the end of each year.

- Write an expression for the amount in the account after:

i 1 year ii 2 years iii 3 years

- The amount in the account after n years can be expressed as:

$$50\,000 \times (1.059)^n + 3000 \times (1.059)^{n-1} + \dots + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$$

This can be written as $50\,000 \times (1.059)^n$, plus the sum of a geometric sequence. Write an expression representing this sum.

- Hence, determine the total value of Iain's savings in 2024, to the nearest dollar.

- 28** Lisa invests \$30 000 at a rate of 1.5% per month compounded monthly. Each month, she withdraws \$600 from her investment after the interest is paid and the balance is reinvested in the account.

- Write an expression for A_1 , the balance of the account after 1 month.
- Write an expression for:

i A_2 ii A_3

- Hence, write an expression for A_n .



- 29** To save up to buy a car, Laura opens a savings account that earns 6% per annum compounded monthly.
She initially deposits \$1400 when she opens the account at the beginning of the month, and then deposits \$165 at the end of every month.
- a** The amount in the account after n months can be expressed as the n th term of a geometric sequence plus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months.
 - b** Hence, determine the amount Laura has saved after 3 years.
- 30** Rochelle invests \$190 000 at a rate of 7% per annum compounded annually, and wants to work out how much she can withdraw each year to ensure the investment lasts 20 years.
- a** Write an expression for A_1 , the balance of the account after 1 year. Use x to represent the amount to be withdrawn each year.
 - b** Write an expression for:
 - i** A_2
 - ii** A_3
 - c** Hence, write an expression for A_n .
 - d** Hence, determine Rochelle's annual withdrawal amount.
- 31** Mario invests \$110 000 at a rate of 16% per annum compounded quarterly and wants the investment to last 25 years.
- a** The amount in the account after n quarters can be expressed as the n th term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n quarters. Use x to represent the amount to be withdrawn each quarter.
 - b** Hence, determine Mario's quarterly withdrawal amount.
- 32** Emma invests \$190 000 at a rate of 3% per annum compounded monthly, and wants to work out how much she can withdraw each month to ensure the investment lasts 30 years.
- a** The amount in the account after n months can be expressed as the n th term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months. Use x to represent the amount to be withdrawn each month.
 - b** Hence, determine Emma's monthly withdrawal amount.